

Mathieu Lacage

Software Engineer · Systems & Machine Learning

📍 Antibes, France ✉ mathieu.lacage@cutebugs.net 🌐 cutebugs.net
🐙 [mathieu-lacage](https://github.com/mathieu-lacage) 🎓 Scholar 🌐 [mathieulacage](https://www.linkedin.com/in/mathieulacage)

Hands-on engineer: co-created ns-3 (2,000+ papers/year), 12 years as CTO scaling a SaaS to 100M messages/day and €4M ARR, now building small language models at INRIA.

Skills

- Languages: C, C++, Python, Go, SQL
- Infrastructure: Podman, MySQL, Ansible, uWSGI, Prometheus, Grafana
- Machine learning: PyTorch
- Networking: InfiniBand, Wi-Fi MAC, IP, TCP

Research Engineer, INRIA

2025 – Present

Researching domain-specific SLMs (Small Language Models) under 1B parameters

- Domain-specific tokenizer adaptation strategies that do not require re-training models from scratch
- Dataset resampling to speed up pre-training loss convergence
- Training convergence stability of MoE models

Select software projects

- 1B GPT-2-style model with mixed-precision pretraining and SFT in PyTorch on data-parallel 2×H100, grid'5000
- Roofline-based inference timing model, to create interactive visualizations of cost, pricing, and margin models
- Podman-based single-GPU SLM training framework
- [naw](#): compressed on-disk streaming of training metrics for long-duration runs

Architect, Hive

2023 – 2025

Decentralized cloud storage on top of a Go-based libp2p network

Scaled storage capacity (100×) and maintained data availability above 99.99%

- Managed the backend team (4)
- Metric collection, analysis, monitoring, and alerting spanning centralized infrastructure and our p2p network
- Debugged and fixed network and IO problems as our production load scaled
- Built Python-based development, testing, and debugging tools
- Planned and managed component-level testing strategies
- Designed forward error correction policies around dynamic decisions to maximize data availability and minimize storage use

Conversational customer care solution for large enterprises, such as LVMH, SNCF, FNAC, Carrefour

As CTO, I

- Made sure our customers' data stayed safe via at-rest encryption and centralized access control
- Scaled our code and infrastructure to handle peaks of 100M messages per day
- Built and led the engineering team from 0 to 10
- Mentored and directly managed the engineering manager and product director

As co-founder, I focused on

- Delivering same-day fixes to customer issues
- Building prototypes to explore new business opportunities

Select projects

- Pub/Sub streaming WebSocket server that scales to 10K concurrent connections per CPU
- Zero-downtime scale up of a single database with a single table to partitioned databases per customer, 250 tables, 1TB
- Centralized access control for SSH via Google SSO and SSH certificates
- A security policy that spans infrastructure and application-level code
- Metabase dashboards consolidating CRM, volume and billing data to track MRR and ARR metrics

Software Lead, ns-3, INRIA

2005 – 2011

ns-3, the de facto open-source network simulator for network research used in 2,000+ papers/year

Co-led its development with the University of Washington and Georgia Tech teams.

- Designed and implemented core APIs: object model, network packets, event scheduler
- Implemented models for UDP/IP/ICMP, MAC/PHY Wi-Fi network protocols
- Integrated unmodified real-world network protocol implementations
- Recruited and managed a local development team (5)
- Relocated to the University of Washington for 10 months to initiate collaboration with the US team
- Evangelized use of ns-3 within other research institutes via presentations and seminars
- Published 7 papers on ns-3 (4 as main author and 3 as co-author)

Software Engineer, INRIA

2003 – 2005

Designed and built software for network research teams: event-driven simulation, distributed experiment management, Linux kernel Wi-Fi drivers.

Provided software mentoring to research projects involved in bio-reactor chemical reaction control and medical image analysis.

Software Engineer, Sigma-Designs

2001 – 2003

Video decompression chips

Implemented DVD navigation control software for chips sold to OEMs to build consumer DVD players.

Education

2006 – 2010	Ph.D. at University of Nice, <i>Experimentation Tools for Networking Research</i> , under the supervision of Walid Dabbous
1998 – 2001	Engineer at Telecom ParisTech (ENST), Software Engineering, Networking, Micro-Electronics